Can AI Bridge the Gender Gap in Sports? Evidence from YouTube and Online Experiments

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Abstract

Women's sports receive far less coverage than men's. Generative artificial intelligence (AI) could expand their social media exposure at scale. We investigate whether AI commentary can narrow the gender gap in engagement with football highlights. On the official YouTube channels of 54 European clubs, we document that, within each channel, women's fixtures receive fewer and later highlights, which draw 89% fewer views. Although we find no evidence that women's highlights carry commentary less often, their commentary is more often female-voiced and less excited. To test whether commentary matters, we randomize it on the same men's and women's goal-highlight clips in two preregistered experiments with 1,298 U.K. residents who watch football regularly. Excited commentary raises engagement, voice gender matters far less, and AI commentary falls short of the original human commentary by less than half the excitement effect. However, neither AI nor excited commentary bridges the gender gap. Exploratory analyses indicate that the gap is concentrated among male participants and those who rarely watch women's football, whereas commentary moves engagement in every group. AI commentary could help extend coverage of women's football, but building a lasting audience will likely require measures beyond the commentary itself.

Keywords: gender gap, sports media, AI-generated content, synthetic voice, online engagement, randomized experiment

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1 Introduction

Women's sports have gained prominence, yet they still receive under 10% of the coverage on televised sports news and highlight shows, after three decades of little change (Cooky et al. 2026). This disparity can perpetuate itself, because how often and how women's sports are covered shapes how audiences value them (Fink 2015).

Social media have expanded access to sports content and the visibility of women's sports (Cooky et al. 2026), and short-form highlights on platforms such as YouTube carry sports content beyond broadcast coverage. However, their visibility depends on algorithmic engagement signals (Petersen-Wagner et al. 2026), and recommenders built on other users' choices shift market share toward items that are already popular (Lee and Hosanagar 2019). Lower production quality may suppress engagement and revenue, reinforcing a cycle of underinvestment that widens the gap in coverage and in the value of women's sports media rights (Musto et al. 2017, Vonwiller et al. 2025).

Commentary is one production mechanism that may shape engagement with sports highlights. Video ads that brands post on YouTube are shared more when they evoke positive emotions such as excitement (Tellis et al. 2019). Televised sports news and highlight shows often present women's sports with less vocal animation and excitement, a pattern Musto et al. (2017) describe as "gender-bland sexism." Generative AI commentary could add excitement to highlights at scale and extend media coverage to fixtures that now receive little.

Little is known about whether generative AI can narrow the gender gap in sports media coverage and engagement. We study AI commentary on YouTube highlights of club football and ask three questions. First, can AI commentary substitute for the original human commentary, and does replacing it narrow the engagement gap between women's and men's football? Second, does the gender of the commentary voice affect engagement? Third, does excited commentary raise engagement, and more so for women's football?

Answering these questions separates two explanations for the gap. One locates it in production: women's sports are presented with less excitement and fewer production resources (Musto et al. 2017, Fink 2015), which lower production costs could address. The other locates it in the audience: responses to the same performance can vary with who responds and whether a gender cue is present (Aguiar 2026, Chen et al. 2023, Gomez-Gonzalez et al. 2024, Rogers 2020). Field data cannot tell the two apart, because production and audience change together. An experiment can: generative AI lets us vary the commentary while the footage and the audience stay fixed. This design yields two contributions. For research on gender gaps in sports media, we test whether the gap moves with the commentary. For research on generative AI in content production, we benchmark unlabeled AI commentary against the original human commentary on the same footage. Earlier work finds that AI beats the average human piece but not the best one (Heitmann et al. 2026); other studies hold content fixed while varying the AI label (Luo et al. 2019, Carney et al. 2026) or compare written match reports blind (Duncan et al. 2025).

We combine an archival study (Study 1) with two experiments (Studies 2 and 3). Study 1 covers the 15,707 club-fixtures that 54 clubs in the top five European leagues played over six seasons, 2018/19 to 2023/24. Over the same seasons, the clubs posted 9,296 first-team highlight videos with retrievable audio on their official YouTube channels. From each video's audio we measure whether it carries commentary

and, when it does, the gender of the voice, its excitement, and the number of player names mentioned per minute. In the two preregistered within-subjects experiments, 1,298 U.K. residents who watch football regularly rate the same eight goal-highlight clips, four men's and four women's, with the commentary assigned at random. Study 2 replaces the original human commentary with AI commentary. Study 3 uses AI commentary throughout and varies, in a 2×2 design, whether it is excited or monotone and whether its voice is male or female. We calibrate the AI commentary to Study 1's commentary corpus, and the same audio instruments confirm the manipulations.

In the field, within the same club channel, a women's fixture is 31% less likely to receive a highlight, and its first highlight is posted 51% later than the men's. A women's highlight draws 89% fewer views yet collects 48% more likes per 1,000 views, and men's and women's highlights carry commentary at rates the sample cannot separate. In the experiments, excited commentary raises engagement by 0.82 points on the 1–7 index. AI commentary falls short of the original human commentary by 0.35 to 0.38 points, less than half the excitement effect. Neither AI nor excited commentary closes the gap between women's and men's clips, which persists at 0.69 points under human commentary and 0.51 points with AI commentary throughout. The gap is concentrated among male participants and among those who rarely watch women's football, whereas the commentary moves engagement in every group. Changing the commentary thus moves the level of engagement, not the gap, which appears to begin in the audience.

2 Archival Evidence: The Gender Gap within Club Channels (Study 1)

2.1 Data

European football clubs post match highlight videos on their official YouTube channels, typically within a day of the match and often with commentary. Our data come from 54 clubs that run one official channel for both their men's and women's teams. A club-season enters the sample when the men's team plays in one of the top five European leagues (England, Spain, Italy, Germany, and France) and the women's team in the same country's top division. Of the 324 club-seasons from 2018/19 to 2023/24, 231 qualify (Online Appendix Tables A2 and A3).[1] For these club-seasons we record, mainly from FBref (fbref.com), the full schedules of these domestic leagues and European club competitions and keep every completed fixture of the 54 clubs: 15,707 club-fixtures. From the channels we collect every first-team match highlight for these seasons and keep the 9,296 with retrievable audio; 22.6% are women's-team highlights. For each video we observe its public metadata (title, tags, duration, upload time, and view, like, and comment counts) together with its audio. We link 6,803 of these videos to the fixtures they cover, and 6,163 club-fixtures receive at least one highlight.

We measure eight outcomes in three categories. *Audience response*: for each video we observe its views and its likes per 1,000 views. *Match coverage*: for each club-fixture we observe whether it receives a highlight and, when it does, how many hours pass from the scheduled start of the match to the first one. *Highlight production*: we measure the other four outcomes from the audio. We transcribe every video with Whisper (Radford et al. 2023). We code commentary as present when the transcript has at least 0.2 words per second and less than 4% first-person words. We measure the other three outcomes on videos with commentary. We

classify the commentator's voice as male or female with a speaker-gender model (Burkhardt et al. 2023). We score the arousal of every speech segment with a speech-emotion model (Wagner et al. 2023). We define excitement as the 90th percentile of a video's segment scores. A zero-shot named-entity model, GLiNER (Zaratiana et al. 2024), tags the person names in each transcript. We merge mentions by surname and count the distinct names per minute.

2.2 Specification

For each outcome we estimate the following specification:

$$Y_{ic} = \alpha + \beta \,\text{Women}_i + \mathbf{X}_i\gamma + \delta_c + \epsilon_{ic}, \tag{1}$$

where Y_{ic} is the outcome for unit i on channel c, Women_i equals one for the women's team, $\mathbf{X}_i$ is a vector of controls, and δ_c denotes channel fixed effects. The unit is a video, or a club-fixture for the two coverage outcomes. The audience-response and delay regressions control for match characteristics; Online Appendix Table A4 lists every control. The audience-response and coverage regressions also include season and competition-type fixed effects; the four production regressions replace these and δ_c with channel×season fixed effects. The coefficient of interest, β, is the difference in the outcome between women's and men's football within the same channel. Standard errors are clustered by channel.

2.3 Results

Figure 1 displays the eight estimates of β.

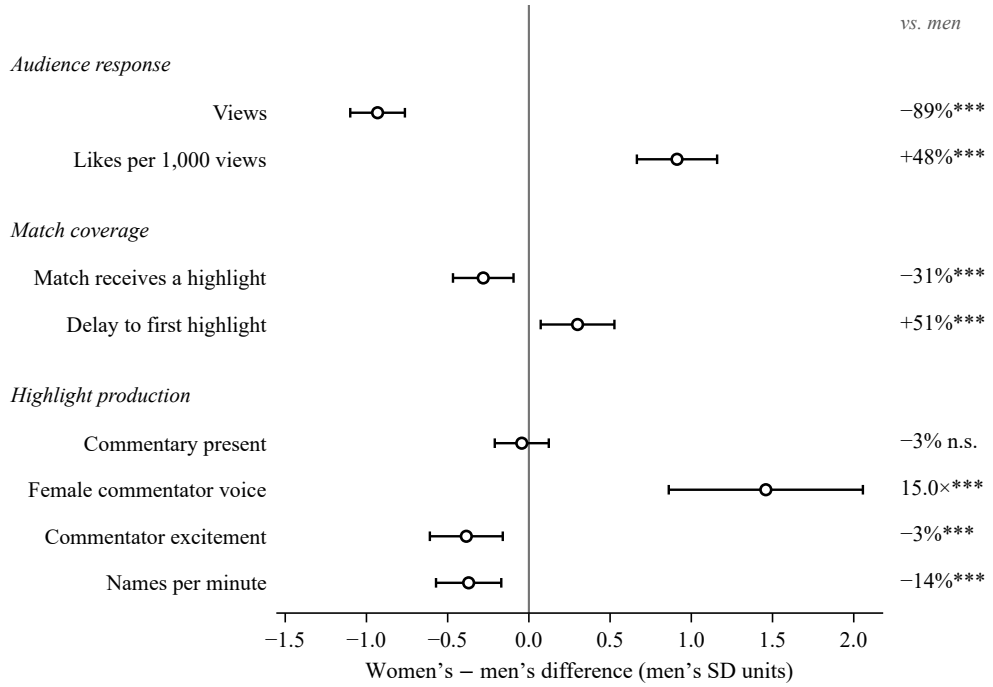
Audience response. Within the same channel and conditional on match characteristics, a women's highlight attracts 89% fewer views than a men's highlight, yet it collects 48% more likes per 1,000 views. The median views are 115,400 for men's highlights and 5,700 for women's.

Match coverage. Overall, 45% of men's fixtures and 30% of women's receive a highlight. Within the same channel, a women's fixture is 31% less likely to receive one at all, and when it does, its first highlight is posted 51% later. The median delays are 8.6 hours after the scheduled start for men's fixtures and 22.0 for women's.

Highlight production. Two-thirds of men's highlights carry commentary, a share the sample cannot separate from the women's within the same channel. The commentary itself differs on three dimensions. The commentary voice is female 15 times as often on women's highlights (13.5% of women's highlights with commentary against 0.9% of men's). The commentator's excitement is lower by about an eighth of the excited-to-monotone range that Study 3 spans, a gap that holds within male-voiced commentary. Commentary on women's highlights names 14% fewer players per minute.

Archival data cannot disentangle audience tastes, quality cues, and amplification: clubs choose video production in anticipation of demand, and an unobserved algorithm allocates exposure. Nor can the data tell whether women's football is less liked or simply less seen: likes per 1,000 views run higher on women's highlights. The gap the data do measure comes earlier, in whether a club posts a highlight and in how many views it draws. The commentary differs, but it explains little of the views gap: adding excitement and names

Figure 1. Study 1: The Gender Gap within Club Channels



Note. Markers are β from Equation 1, one regression per outcome (Online Appendix Table A4), divided by the men's standard deviation (SD); views and delay are in logs. Whiskers are 95% confidence intervals clustered by channel. The *vs. men* column restates β relative to the men's level (logs: $e^{\beta} - 1$; female voice: ratio of shares). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, two-sided.

per minute to the views regression absorbs 3.3% of it. If the shortfall in coverage reflects the cost of producing a highlight with commentary, AI commentary could lower that cost but would help only if audiences respond to it as they do to human commentary. We therefore turn to two experiments that hold the footage and the audience fixed.

3 Experimental Evidence: Same Footage, Randomized Commentary

Study 2 replaces human commentary with AI commentary; Study 3 varies the excitement and voice gender of AI commentary. Each study asks whether the commentary changes engagement and whether it narrows the gender gap, the difference in engagement between women's and men's clips (women's − men's). Section 3.1 describes the stimuli, design, sample, and estimation the two studies share; Sections 3.2 and 3.3 report the results, and Section 3.4 examines how they vary across participants.

3.1 Overview of the Experiments

3.1.1 Stimuli

The footage is eight official goal-highlight clips, 13 to 37 seconds long, from the 2025/26 UEFA Champions League and Women's Champions League, all from non-U.K. clubs. Each sport contributes one goal of each type: a team move, a solo goal, a free kick, and a long shot. Each clip exists in five versions on the same

footage and stadium sound: the original human commentary and four AI commentary versions, one for each combination of voice gender and excitement. The five versions play at similar loudness, the excited ones about 1 dB louder, so excited and monotone differ mainly in expression. We produced the AI versions ourselves: we drafted the scripts with large language models, fact-checked them against the match record, and calibrated them to Study 1's commentary corpus. We then synthesized the voices and mixed them onto the original stadium sound. The four AI versions of a clip share the script, timing, and mix and differ only in the synthesized voice. Study 1's speaker-gender and speech-emotion instruments (Burkhardt et al. 2023, Wagner et al. 2023) confirm both manipulations. At either level of excitement, the female voice registers as slightly less excited than the male voice, so part of the voice contrast in Section 3.3 may reflect excitement rather than gender. On the speech-emotion instrument, the excited AI versions are at least as excited as the human commentary, and the human commentary is matched in excitement across the two sports. The human commentary is therefore a best-case benchmark: professional commentary without the excitement gap Study 1 finds in the field.

3.1.2 Design and Randomization

Both studies use the same within-subjects design. Every participant watches each of the same eight clips once, in random order, and rates both sports under every commentary version, so the design holds the audience fixed as well as the footage. Each participant's own women's−men's difference also becomes an observation for Section 3.4. The design's cost is that a participant hears several versions of similar content, so Sections 3.2 and 3.3 also check the commentary effects on participants' first clips, a between-subjects comparison. In Study 2, two randomly chosen clips per sport carry the original human commentary and the other two carry the AI version with the excited male voice, so both AI and human commentary are excited. In Study 3, all eight clips carry AI commentary; within each sport, the four clips cover the four voice-by-excitement conditions in a Latin-square rotation, so every participant hears every condition in both sports. Across participants, each clip appears equally often in each version. Participants learn only that they will watch and rate football highlight clips. We mention neither the commentary nor the use of AI before the debrief, because labeling content as AI-made can depress responses even when the content is identical (Luo et al. 2019, Carney et al. 2026).

3.1.3 Sample

We recruited 660 participants per study on Prolific in August 2026, for $2.50 each; no participant took part in both studies. Prolific's prescreens, based on stored profile answers rather than questions asked at study time, admitted only U.K. residents who watch football regularly, speak fluent English, and hold an approval rate of at least 95%. The study also required a desktop computer with audio. An audio check, in which participants report the number spoken in a short recording, precedes the first clip and is repeated after the last. We keep participants who completed the study, passed the second audio check, and did not give identical answers on all 24 ratings (three items × eight clips). This leaves 643 participants in Study 2 (mean age 38.9; 32% female) and 655 in Study 3 (mean age 40.3; 35% female); age and gender come from Prolific's profile records. Online Appendix A reports the exclusion funnel. The protocol was approved by the Human

Subjects Committee of the Faculty of Economics, Business Administration and Information Technology at the University of Zurich (OEC IRB #2025-116), and predictions, design, sample size, exclusion rules, and analysis plan were preregistered at OSF before data collection began.[2]

3.1.4 Outcomes and Estimation

After each clip, participants rate on seven-point scales how much they enjoyed it, how likely they are to share it, and how likely they would be to "watch more highlights like this one." We adapt the three items from Tonietto and Barasch (2021), Berger and Milkman (2012), and Zhang et al. (2025), respectively. We average them into an engagement index on the 1–7 scale, the primary outcome. For Study 2 we estimate, on the 5,144 clip-level observations (643 participants × 8 clips),

$$Y_{ij} = \alpha + \beta_1 \text{Women}_j + \beta_2 \text{AI}_{ij} + \beta_3 (\text{Women}_j \times \text{AI}_{ij}) + \mathbf{X}_i \gamma + \delta_g + \epsilon_{ij}, \tag{2}$$

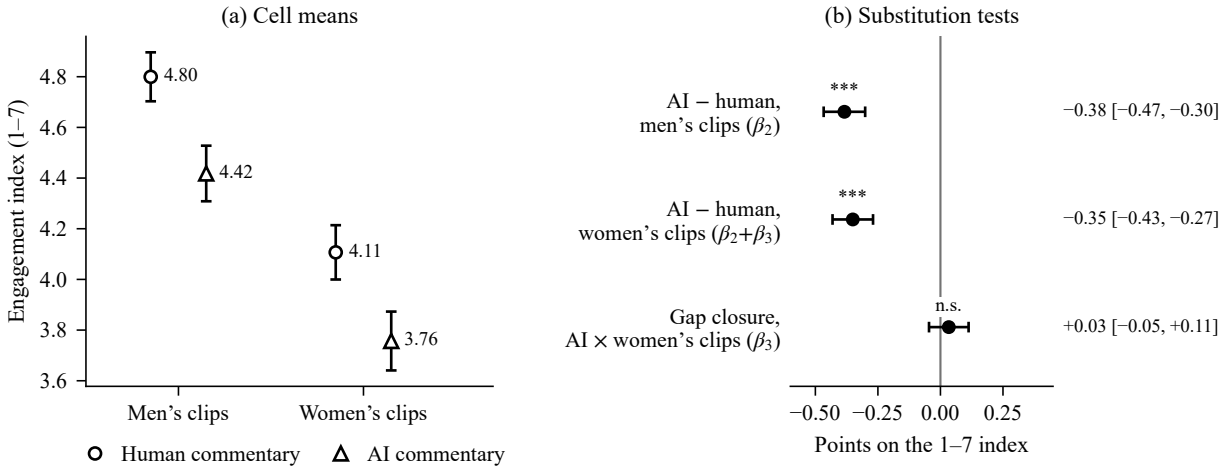
where Y_{ij} is participant i's engagement index for clip j, Women_j indicates a women's clip, and AI_{ij} indicates AI rather than human commentary. The vector $\mathbf{X}_i$ collects participant gender, age, and two baseline items on how often the participant watches men's and women's football, and δ_g denotes goal-type fixed effects. Standard errors are clustered by participant, because each participant contributes eight ratings. The coefficients of interest are β_2 and $\beta_2 + \beta_3$, the AI−human differences on men's and women's clips, and β_3, the change in the gender gap when AI commentary replaces human commentary. We test the two differences one-sided against a substitution margin of −0.25 points (Lakens 2017), the smallest difference we judged would change the decision to replace human commentary; the gap-closure prediction is $\beta_3 > 0$.

Study 3 keeps the same specification and replaces the AI indicator with the two factors varied in the AI commentary, Excited_{ij} (excited versus monotone) and Female_{ij} (female versus male voice). These enter Equation 2 in place of AI_{ij}, together with all their interactions with each other and with the sport factor (5,240 observations, 655 participants × 8 clips). We code each factor as $-1/2$ and $+1/2$ rather than 0 and 1. Each main effect is then the factor's effect averaged over the other two factors, and each two-way interaction is averaged over the third, rather than at the reference level of the other factors. Four coefficients answer the questions: excitement (β_2), voice (β_3), excitement × voice (β_4), and excitement × women's clips (β_5); we make no predictions for the remaining interactions. The predictions are $\beta_2 > 0$ and $\beta_5 > 0$, that excited commentary raises engagement and narrows the gender gap. For the voice effects β_3 and β_4 we make no directional prediction; we report 95% confidence intervals and run two one-sided tests against ±0.25 points. With 80% power, the studies can detect effects of 0.06 to 0.13 points, well below 0.25 points.

3.2 Study 2: AI versus Human Commentary

Figure 2 reports the results. Panel (a) shows two patterns: every AI mean lies below its human counterpart, and the gender gap is nearly the same under AI and human commentary. The estimates in panel (b) put the AI−human difference at −0.38 points $[-0.47, -0.30]$ on men's clips and −0.35 points $[-0.43, -0.27]$ on women's. On participants' first clips, the between-subjects check of Section 3.1.2, the difference is −0.35 ($p < 0.01$). Both confidence intervals lie entirely below the −0.25 margin: AI commentary falls short of

Figure 2. Study 2: AI versus Human Commentary



Note. Panel (b) estimates Equation 2 with controls and goal-type fixed effects. Whiskers are 95% confidence intervals clustered by participant. $N = 643$ participants. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, two-sided.

human commentary by more than the substitution margin.

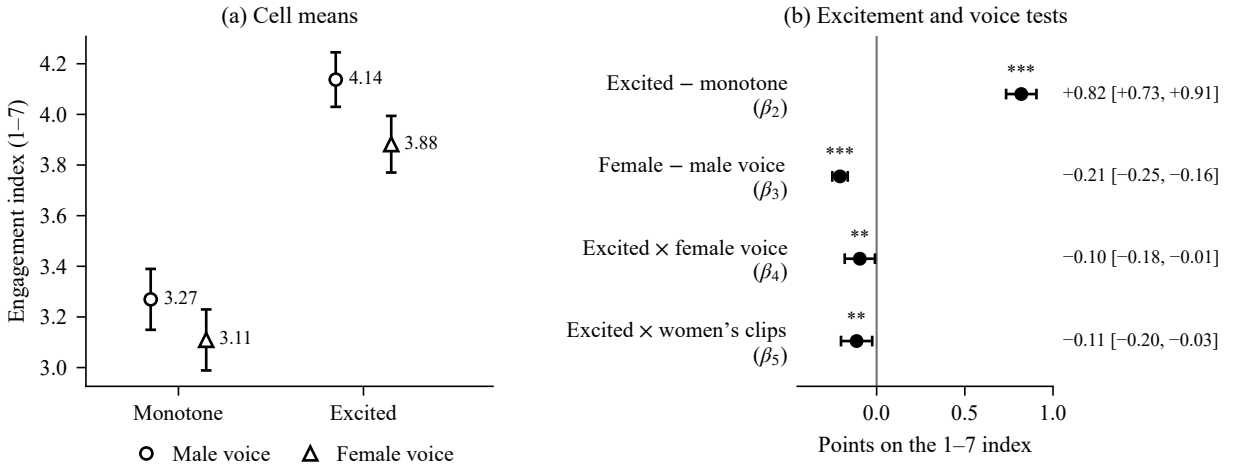
We turn to the gender gap. Under human commentary, women's clips score 0.69 points below men's $[-0.78, -0.61]$: the gap persists when the same participants rate both sports. Replacing human commentary with AI commentary does not close it. The change in the gap under AI commentary is +0.03 points $[-0.05, +0.11]$, which the sample cannot separate from zero, so the gap-closure prediction finds no support.

3.3 Study 3: Excitement and Voice Gender

Figure 3 reports the results. Panel (a) orders the four conditions: both excited conditions lie well above both monotone conditions, and within each level of excitement the male voice lies slightly above the female voice. The estimates in panel (b) put the excitement effect at +0.82 points $[+0.73, +0.91]$, more than twice Study 2's AI−human difference. On participants' first clips, the between-subjects check of Section 3.1.2, it is +0.49 ($p < 0.01$). The female voice scores 0.21 points below the male voice $[-0.25, -0.16]$, a quarter of the excitement effect. The interval reaches the −0.25 bound, and the one-sided test rules out −0.25 only narrowly ($p = 0.028$). The two interact: the excitement effect is 0.10 points smaller with a female voice $[-0.18, -0.01]$, well inside the ±0.25 bounds, and remains large with either voice. Excitement matters far more than voice gender.

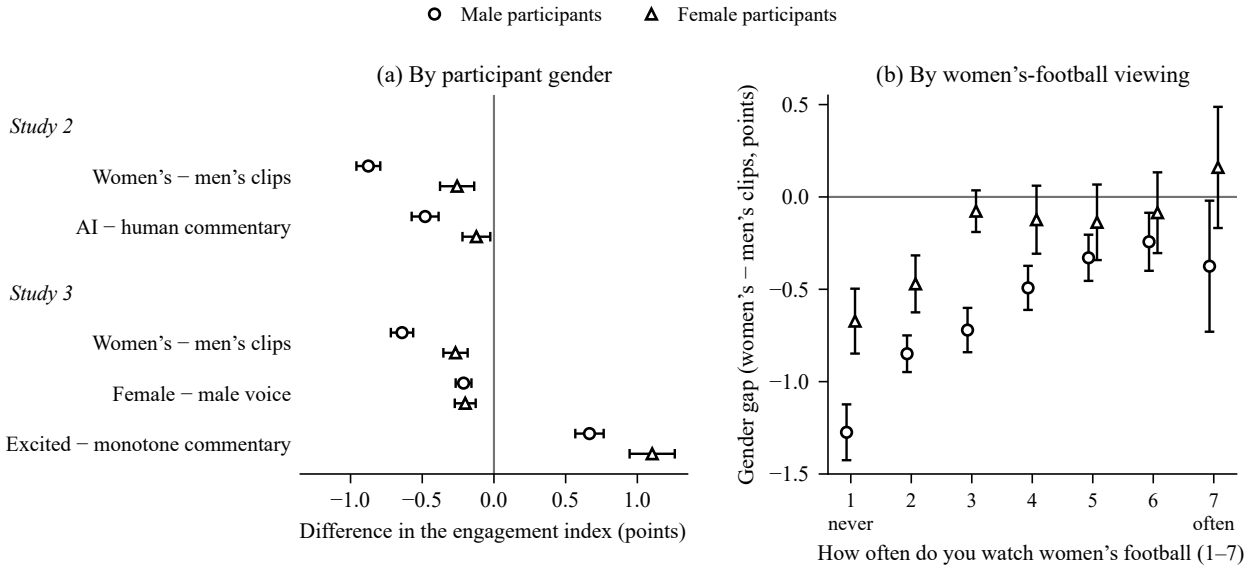
We turn again to the gender gap. With AI commentary on every clip and the four voice-by-excitement conditions balanced across sports by design, women's clips still score 0.51 points below men's $[-0.57, -0.45]$. The prediction that excited commentary narrows the gap fails in the opposite direction: the excitement effect is 0.11 points smaller on women's clips $[-0.20, -0.03]$, though it remains large for both sports. Across the two studies, replacing human commentary with AI commentary does not close the gender gap, and neither does excited commentary.

Figure 3. Study 3: Excitement and Voice Gender



Note. Panel (a) pools over sport; panel (b) estimates the Study 3 specification (factors coded ±1/2, so each estimate is averaged over the remaining factors; controls, goal-type fixed effects). Whiskers are 95% confidence intervals clustered by participant. N = 655 participants. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, two-sided.

Figure 4. Exploratory Evidence: Heterogeneity across Participants



Note. Markers are means of participant-level contrasts, the first condition minus the second, without controls; whiskers are 95% confidence intervals. Panel (b) pools Studies 2 and 3 (N = 1,297) and splits the women's−men's contrast by participant gender and women's-football viewing; the smallest group holds 22 participants.

3.4 Exploratory Evidence: Heterogeneity across Participants

Figure 4 splits five contrasts of Sections 3.2 and 3.3 by participant gender (panel (a)) and the gap by how often participants watch women's football (panel (b)). The gap is concentrated among male participants and among those who rarely watch women's football. It is more than three times as large among male as among

female participants in Study 2 (−0.88 against −0.26) and more than twice as large in Study 3 (−0.64 against −0.27). Within each gender, the gap narrows as women's-football viewing rises. Among participants who never watch women's football, it is −1.27 for male and −0.67 for female participants. Each additional scale point on the viewing item is associated with a smaller gap, by 0.10 to 0.26 points (each $p < 0.001$), holding men's-football viewing and age constant.

Every commentary effect appears among both male and female participants. The AI−human difference is larger among male participants (−0.48 against −0.12) and grows with men's-football viewing (−0.09 per scale point). The voice contrast differs by only 0.01 points between male and female participants (−0.21 and −0.20) and changes by at most 0.03 points per scale point of either viewing item. The excitement effect is larger among female than among male participants (+1.10 against +0.67). Overall, the gap depends on who watches, whereas the commentary moves engagement in every group.

4 Discussion and Conclusion

Study 1 documents, within the same club channels, the gap that research on sports media finds across outlets (Cooky et al. 2026): women's fixtures receive fewer and later highlights, which draw 89% fewer views. Studies 2 and 3 vary only the commentary, which moves engagement with both sports, and AI commentary falls short of the original human commentary. Yet the gap persists in both studies, so AI commentary does not bridge it.

Taken together, the studies suggest that the gap lies in the audience rather than in the commentary. Study 1 is consistent with this reading: if the audience that women's highlights reach is self-selected, those highlights should draw more likes per view, as they do. The same pattern appears in other ratings: for example, crowds, but not critics, rate films lower with a female lead (Aguiar 2026). Participants also rate men's goals above women's when they can see the players' gender, and by less when blurring hides it (Gomez-Gonzalez et al. 2024).

The results have implications for two decisions clubs face: whether to replace human commentary with AI commentary, and how to produce AI commentary where they use it. AI commentary scores below the original human commentary by more than the substitution margin, so clubs that already have commentary of that quality should keep it. Where a highlight has no commentary, as a third of men's highlights in our data do not, the alternative to AI commentary is no commentary, a comparison our experiments do not make. For the second decision, excitement matters most. Excited AI commentary scores above monotone AI commentary by more than twice the AI−human difference, so AI commentary should be excited. The voice matters far less: the female voice scores below the male voice by a quarter of the excitement effect. Clubs use a female voice far more often on women's highlights, but because the gap persists in Study 3, where voices are balanced across sports, that practice is unlikely to explain the gap.

The commentary differences explain little of the views gap, and platforms shape views through exposure we do not observe. Women's highlights draw more likes per view, so the audience they reach appears to respond more strongly. The experiments provide a note of caution: the gap is concentrated among participants who rarely watch women's football, so wider exposure may reach an audience that responds less. Platforms

could raise the exposure of women's highlights and measure whether the higher response per view persists as exposure widens. Leagues could require a highlight of every women's fixture, posted as promptly as men's highlights. We do not test either step and leave their effectiveness to future research.

More broadly, with generative models, marketing text and video can now be produced at a fraction of the cost of human production (Reisenbichler et al. 2022, Kapoor and Kumar 2025). Whether lower costs help content that now receives little coverage or no commentary depends on where its shortfall lies. Lowering the cost of an input can move a shortfall that comes from the input, but not one that comes from the audience. Our experiments estimate the response to unlabeled commentary: participants were not told, before the debrief, that any commentary was AI-made, and 9% in Study 2 mentioned AI unprompted. The European Union's AI Act requires providers to mark synthetic audio and video as artificially generated, in a machine-readable format.[3] A label that audiences see may reduce the response further (Luo et al. 2019, Carney et al. 2026); estimating that effect for commentary requires randomizing the label.

Our study has four limitations. First, the experimental evidence comes from one football audience, U.K. residents who watch football regularly, rating goals from the Champions League and the Women's Champions League, and from AI commentary made with current tools. Other sports, audiences, and AI systems may yield different results, which replications could test. Second, the reading that the gap lies in the audience rests on exploratory analyses; an experiment that varies participants' exposure to women's football could test it directly. Third, the experiments measure stated engagement with a highlight clip, not whether commentary that raises engagement also raises demand for women's football over time. Fourth, the archival data cannot tell why only 45% of men's fixtures and 30% of women's receive a highlight. The shortfall may reflect the cost of producing a highlight with commentary, but clubs also post highlights without it. A field experiment that publishes highlights, with and without AI commentary, for fixtures that now receive none would address the last two. Such an experiment would show whether those fixtures draw an audience, whether AI commentary adds to it, and whether that audience returns, with Study 1 as the baseline.

Notes

[1] The sample ends with 2023/24 because the Women's Super League (WSL) moved women's highlights to a central league channel in 2024/25, after which club-channel coverage no longer reveals clubs' choices.

[2] Preregistration: https://osf.io/p7hw2/?view_only=615795b079bd45a6984ea885f4bd0dac.

[3] Regulation (EU) 2024/1689 (Artificial Intelligence Act), Article 50; http://data.europa.eu/eli/reg/2024/1689/oj.

Declarations

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authors declare no other competing interests.

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Online Appendix

A Experimental Samples

Table A1 traces each study from Prolific submission to the analysis sample; the final samples of 643 and 655 exceed the registered target of 550 per study. Study 2 ran on August 3, 2026, and Study 3 on August 4; the OSF registration preceded the launch of Study 2.

Table A1. Sample Funnel

	Study 2	Study 3
Prolific submissions	694	696
returned or timed out	−30	−36
screened out at first audio check	−4	0
Approved submissions	660	660
declined consent	−1	−1
clip-playback error	−1	0
identical answers on all 24 ratings	−15	−4
Analysis sample	**643**	**655**

Note. No participant failed the second audio check. By gender, the analysis samples hold 436 male and 206 female participants in Study 2 (one reported neither) and 424 male and 231 female in Study 3.

B Study 1: Sample and Regressions

Table A2. Study 1 Population: Eligible Clubs by Season and Country

Season	England Premier League WSL	Spain La Liga Liga F	Italy Serie A Serie A Femminile	Germany Bundesliga Frauen-Bundesliga	France Ligue 1 D1 Féminine	Total
2018/19	7	10	5	6	6	34
2019/20	8	9	7	6	7	37
2020/21	8	10	7	7	6	38
2021/22	9	12	10	7	6	44
2022/23	10	9	7	8	4	38
2023/24	9	10	6	9	6	40
Club-seasons	51	60	42	43	35	231
Clubs	10	15	10	9	10	54

Note. Cells count clubs whose men's team plays in the country's top division and whose women's team plays in the women's top division (Liga F was Primera División before 2022/23). Four clubs that meet this rule are excluded because they did not post both teams' highlights on one YouTube channel throughout the six seasons: Napoli, Chievo Verona, Tottenham, and Guingamp. The sample ends with 2023/24 because, from 2024/25, England's Women's Super League posted its highlights on a central league channel.

Table A3. Study 1 Sample by Country

	Club-fixtures		Share with a highlight		Highlight videos	
Country	Men	Women	Men	Women	Men	Women
England	2,281	1,142	0.74	0.42	2,788	556
Spain	2,639	1,868	0.24	0.26	1,208	595
Italy	1,853	994	0.76	0.51	1,994	589
Germany	1,683	1,057	0.12	0.08	510	108
France	1,348	842	0.35	0.25	693	255
Total	9,804	5,903	0.45	0.30	7,193	2,103

Note. Club-fixtures are the completed domestic-league and European matches of eligible club-seasons, each counted once per club. Share with a highlight is the share of club-fixtures with at least one highlight on the club's channel; highlight videos are those with retrievable audio. Fixtures come from FBref (fbref.com), Spain's women's top division in 2018/19–2021/22 from BeSoccer (besoccer.com), and missing start times from ESPN (espn.com).

Table A4. Study 1 Regressions: Audience Response, Coverage, and Production

	Sample levels			Women − men difference			
Outcome	Men	Women	SD (men)	Estimate	SE	vs. men	Obs.
Panel A. Audience response							
Views (thousands)[†]	115.4	5.7	2.41	−2.247***	(0.207)	−89%	6,761
Likes per 1,000 views	20.04	28.29	10.57	9.64***	(1.33)	+48%	6,761
Panel B. Match coverage: whether and when							
Match receives a highlight	0.45	0.30	0.50	−0.140***	(0.047)	−31%	15,707
Delay to first highlight (hours)[†]	8.6	22.0	1.37	0.410***	(0.158)	+51%	5,945
Panel C. Highlight production							
Commentary present	0.67	0.55	0.47	−0.021	(0.040)	−3%	9,296
Female commentator voice	0.0090	0.1346	0.09	0.138***	(0.029)	15.0×	5,925
Commentator excitement	0.93	0.90	0.08	−0.029***	(0.009)	−3%	5,925
Names per minute	5.08	4.69	1.94	−0.718***	(0.198)	−14%	5,925

Note. One regression per row (Equation 1 of the paper). Estimate is β, the women's−men's difference within channel, in the outcome's units; *vs. men* restates it relative to the men's level ($e^{\beta} - 1$ for logged outcomes; female voice: ratio of shares). Men and Women are means, or medians for logged outcomes ([†]); SD (men) standardizes Figure 1 of the paper. Fixed effects: channel, season, and competition type (Panels A and B); channel×season (Panel C). Controls: total goals, absolute goal difference, result, home match, and knockout round (views, likes, delay); log duration and log video age (views, likes); weekend start (delay). Samples: linked videos with visible counts (Panel A); covered club-fixtures with a known start time (delay); videos with commentary (last three rows). Coverage is a linear probability model. Within male-voiced commentary, the excitement estimate is −0.027 (SE 0.009). Standard errors are clustered by channel. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, two-sided.